(Please write your Enrollment Number) Enrollment No. 032

End-Term Examination

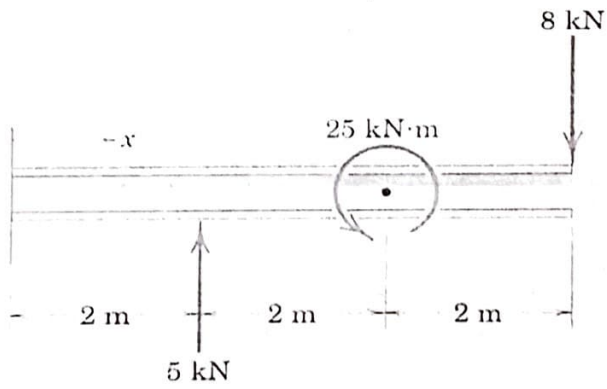
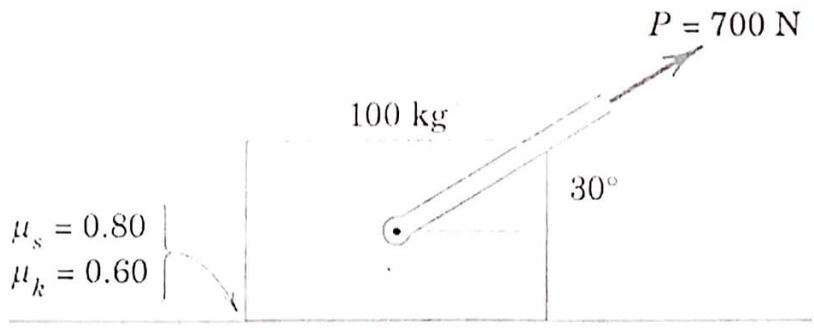
(B.Tech IT/MAE/DMAM-CBCS)(SUBJECTIVE TYPE)

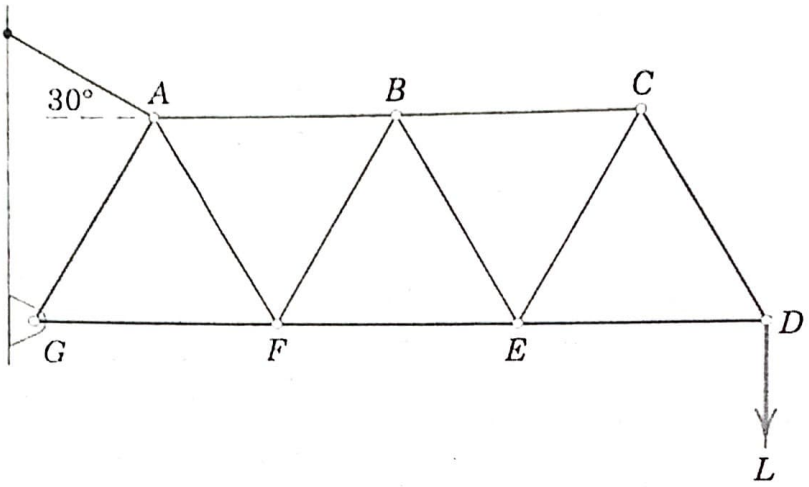
(April, 2022)

Subject Code: BMA 110	Subject: Engineering Mechanics

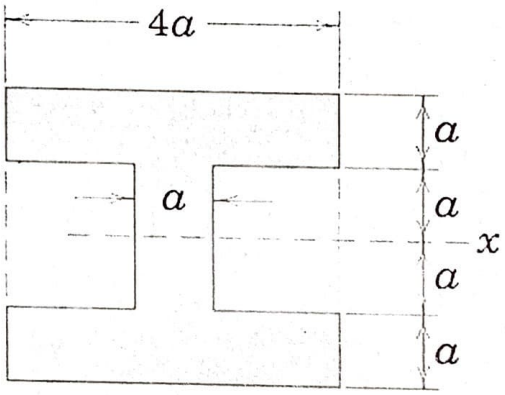
Time : 3 Hours Maximum Marks : 60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1.		**(2.5*8=20)**
	(a) What is principle of transmissibility?	
	(b) Explain cone of friction with the help of a diagram.	
	(c) Explain angle of repose with the help of a diagram.	
	(d) State the theorems of Pappus.	
	(e) Differentiate between Center of Mass and Centroid.	
	(f) Explain parallel axis theorem.	
	(g) What is work-energy principle?	
	(h) What is instantaneous center of zero velocity? How do we find it for a rigid body?	
	UNIT-1	
Q2.	Determine and locate the resultant **R** of the two forces and one couple acting on the I-beam. 	**(10)**
Q3.	The 700-N force is applied to the 100-kg block, which is stationary before the force is applied. Determine the magnitude and direction of the friction force F exerted by the horizontal surface on the block. 	**(10)**
	UNIT-2	
Q4.	Determine the forces in members *BC*, *BE*, and *BF*. The triangles are equilateral.	**(10)**

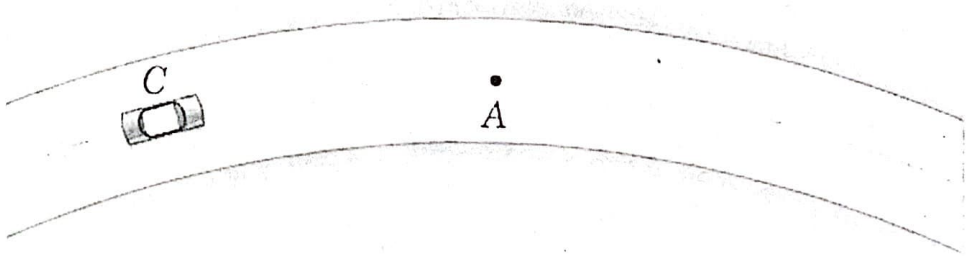


Q5.	Determine the moment of inertia of the shaded area about the x-axis.	(10)

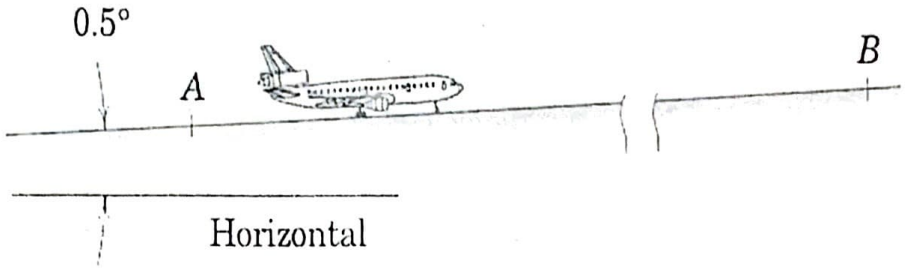


UNIT-3

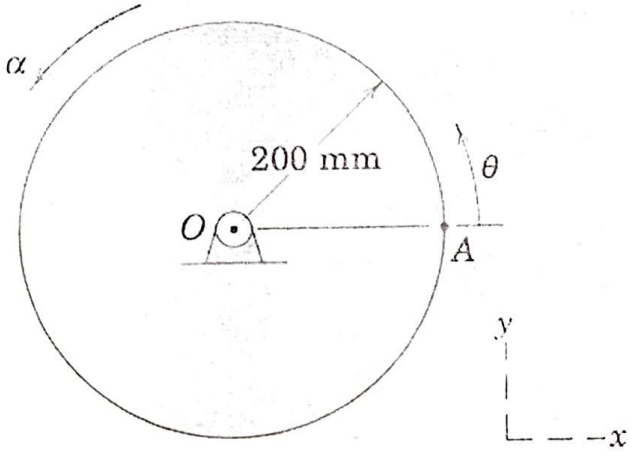
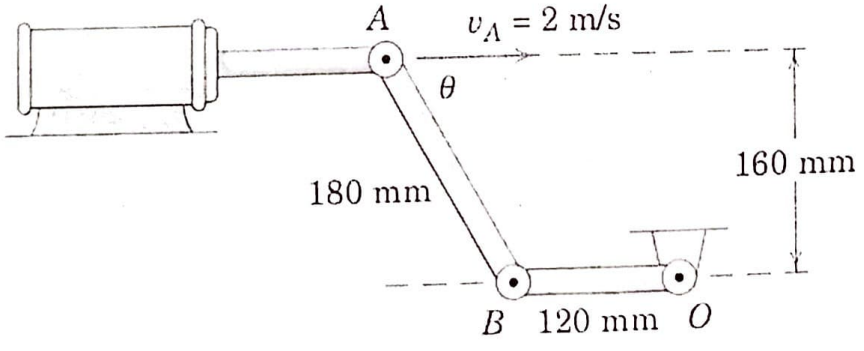
Q6.	The car *C* increases its speed at the constant rate of 1.5 m/s2 as it rounds the curve shown. If the magnitude of the total acceleration of the car is 2.5 m/s2 at the point *A* where the radius of curvature is 200 m, compute the speed *v* of the car at this point.	(10)



Q7.	The 300-Mg jet airliner has three engines, each of which produces a nearly constant thrust of 240 kN during the take-off roll. Determine the length *s* of runway required if the take-off speed is 220 km/h. Compute *s* first for an uphill take-off direction from *A* to *B* and second for a downhill take-off from *B* to *A* on the slightly inclined runway. Neglect air and rolling resistance.	(10)



UNIT-4

Q8.	Point A of the circular disk is at the angular position theta=0 at time t =0. The disk has angular velocity of 0.1 rad/s at t = 0 and subsequently experiences a constant angular acceleration of 2 rad/s^2 . Determine the velocity and acceleration of point A in terms of fixed **i** and **j** unit vectors at time t =1s. 	**(10)**
Q9.	Horizontal motion of the piston rod of the hydraulic cylinder controls the rotation of link OB about O. For the instant represented, vA =2 m/s and OB is horizontal. Determine the angular velocity of OB for this instant. 	**(10)**